

Module 11

Technical Workouts

1. Understand Supervised Learning Fundamentals

- a. What is supervised learning
- b. Learning from labeled data
- c. Classification vs regression
- d. Model training workflow
- e. Practical use cases

2. Understand Regression Algorithms

- a. Linear Regression
- b. Multiple Linear Regression
- c. Regression assumptions
- d. Prediction workflows
- e. Regression applications

3. Understand Classification Algorithms

- a. Logistic Regression
- b. K-Nearest Neighbors (KNN)
- c. Naive Bayes
- d. Classification workflows
- e. Classification use cases

4. Understand Tree-Based Models

- a. Decision Trees
- b. Random Forest
- c. Ensemble learning concepts
- d. Feature importance
- e. Tree-based decision making

5. Understand Boosting Concepts

- a. What is boosting
- b. XGBoost fundamentals
- c. Model improvement concepts
- d. Performance optimization
- e. Practical use cases

6. Understand Model Training Workflows

- a. Data splitting
- b. Training models
- c. Generating predictions
- d. Comparing models
- e. Experiment tracking

7. Understand Feature Importance

- a. Important features
- b. Irrelevant features
- c. Feature impact
- d. Model interpretability
- e. Business insights from models